

1) Set - A
Make groups of 4, then convert to decimal.

0011	1011	1001	1010
↓	↓	↓	↓
3	11	9	10

Subtract 3 from each group.

$3-3=0$ $11-3=8$ $9-3=6$ $10-3=7$

$\Rightarrow (0867)_{10} = (867)_{10}$

Now, convert to base 7.

7	867	Remainder
7	123	6 LSB
7	17	4
7	2	3
	0	2 MSB

$\Rightarrow (2346)_7$

2) $(16)_8 = (001\ 110)_2$
 $(+16) = (0001\ 110)_2$
in 8 bits = $(0000\ 1110)_2$

1's comp. $\Rightarrow (1111\ 0001)_2$
 $+1$
 $(-16) 1111\ 0010$

$(43)_8 = (100\ 011)_2$
 $(+43) = (0100\ 011)_2$
in 8 bits = $(001000\ 11)_2$

$\Rightarrow$

$(+48)_8$	00100011
$+(-16)_8$	11110010
	$\hline 100010101$

discarding carry bit

$\Rightarrow (00010101)_2$

No overflow as final answer is positive.

3) divide $(121231)_5$ by $(132)_5$

$$\begin{array}{r}
 132 \overline{) 121231} \quad 413 \\
 \underline{00243} \\
 132 \\
 \underline{1111} \\
 1001 \\
 \underline{0110}
 \end{array}$$

$\begin{matrix} +5 & +5 \\ 1 & 2 & 1 & 2 & 3 & 1 \end{matrix}$
 $\begin{matrix} +1 & +1 \\ 1 & 1 & 3 & 3 \end{matrix}$

Base 5:

$$132 \times 1 = 132$$

$$132 \times 2 = 314$$

$$132 \times 3 = 1001$$

$$132 \times 4 = 1133$$

$$132 \times 5 = 1320$$

$\therefore$ quotient $\rightarrow (413)_5$

Remainder $\rightarrow (110)_5$

Set - B

1) Make groups of 4, then convert to decimal

$$\begin{array}{cccc} 0101 & 1100 & 1001 & 1011 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 5 & 12 & 9 & 11 \end{array}$$

Subtract 3 from each group.

$$5-3=2 \quad 12-3=9 \quad 9-3=6 \quad 11-3=8$$

$$\Rightarrow (2968)_{10}$$

Now, convert to base 8

8	2968	remainder
8	371	0 LSR
8	46	3
8	5	6
	0	5 MSB

$$\Rightarrow (5630)_8$$

$$2) (11)_7 = (1 \times 7^0 + 1 \times 7^1) \\ = (8)_{10}$$

$$(8)_{10} = (1000)_2$$

$$(+8)_{10} = (01000)_2$$

$$\text{in 6 bits} = (001000)_2$$

$$1's \text{ comp.} \Rightarrow (110111)_2$$

$$+ 1$$

$$2's \text{ comp.} \Rightarrow 111000$$

$$(23)_7 = (3 \times 7^0 + 2 \times 7^1) \\ = (17)_{10}$$

$$(17)_{10} = (10001)_2$$

$$(+17)_{10} = (010001)_2 \\ \text{in 6 bits}$$

$$\begin{array}{r|l} (+23) & 010001 \\ + (-11) & +111000 \\ \hline & 001001 \end{array}$$

X

discard carry bit

$$\Rightarrow (001001)_2$$

No overflow, answer is positive.

3)

$$\begin{array}{r}
 213 \quad | \quad 1 \quad 2 \quad 3^{+5} \quad 2^{+5} \quad 3 \quad 1 \quad | \quad 313 \\
 \hline
 \leftarrow 1 \quad 1^{+1} \quad 4^{+1} \quad 4 \quad \downarrow \quad | \quad \\
 \hline
 \quad \quad \quad 0 \quad 0 \quad 3 \quad 3 \quad 3 \quad | \quad \\
 \quad \quad \quad \quad \quad 2 \quad 1 \quad 3 \quad \downarrow \quad | \quad \\
 \hline
 \quad \quad \quad \quad \quad 1 \quad 2 \quad 0^{+5} \quad 1^{+5} \quad | \quad \\
 \quad \quad \quad \quad \quad 1 \quad 1 \quad 4^{+1} \quad 4 \quad | \quad \\
 \hline
 \quad \quad \quad \quad \quad 0 \quad 0 \quad 0 \quad 2 \quad | \quad
 \end{array}$$

Base -5

$$213 \times 1 = 213$$

$$213 \times 2 = 431$$

$$213 \times 3 = 1144$$

$$213 \times 4 = 1412$$

$$\text{quotient} = (313)_5$$

$$\text{Remainder} = (2)_5$$